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About This Project

This is my final year honours dissertation project submitted in April 2026 as part of my BSc Business Technology degree at the University of the West of Scotland. The project was awarded First Class Honours. It is the first data analytics project I have built entirely from scratch and it covers the full pipeline, from collecting raw data through to producing business recommendations, using Python in Google Colab.

The project analyses customer satisfaction for Tesco and Sainsburys using 2,000 publicly available Trustpilot reviews. I used VADER for lexicon-based sentiment analysis and trained a Multinomial Naive Bayes classifier to predict sentiment from the review text. The keyword findings were then mapped to the SERVQUAL service quality framework to interpret the results in a business context.

2,000

Reviews Analysed

8

Pipeline Stages

68.5%

Model Accuracy

48

References

1st

Honours Class

Tools and Technologies

Python, pandas, scikit-learn, NLTK, VADER (vaderSentiment), matplotlib, wordcloud, Google Colab, Microsoft Excel

The Eight-Stage NLP Pipeline

Stage 1: Loading the Data

I imported 2,000 Trustpilot reviews from Excel into a pandas DataFrame. The dataset had five columns: rating, review title, review content, company name and publication date. There were 1,000 reviews for Tesco and 1,000 for Sainsburys.

Stage 2: Text Pre-processing

I wrote a clean_text function that merged the review title and body into one field, converted everything to lowercase, stripped out punctuation and special characters using regular expressions, and removed extra whitespace. The cleaned output was saved in a column called clean_review.

Stage 3: VADER Sentiment Analysis

I applied VADER to every review and extracted the compound score. Any score above 0.05 was classified as Positive, below negative 0.05 as Negative, and everything in between as Neutral. This produced sentiment labels for all 2,000 reviews.

Stage 4: Sentiment Summary Tables

I used pandas groupby and size to count positive, negative and neutral reviews for each retailer, then produced summary tables showing absolute counts and percentage distributions.

Stage 5: Machine Learning Classification

I vectorised the cleaned text using TF-IDF and trained a Multinomial Naive Bayes classifier on 80 percent of the data. The remaining 20 percent was used as a test set. I evaluated the model using `accuracy_score` and `classification_report` from scikit-learn.

Stage 6: Confusion Matrix

I generated a confusion matrix to understand how the model classified each sentiment category. This revealed that the model failed entirely on neutral reviews due to extreme class imbalance. Only 95 of the 2,000 reviews were neutral, which was not enough for the classifier to learn that class.

Stage 7: Word Clouds

I created word clouds for Sainsburys reviews, Tesco reviews and all reviews combined using the wordcloud library. I built a custom stop word list to remove store names and common filler words that would otherwise dominate the output.

Stage 8: Keyword Frequency Analysis

I used Python's Counter object to extract the top 15 keywords from Sainsburys reviews and from the negative review corpus. These keywords were then mapped to the five SERVQUAL dimensions to interpret which service quality areas were driving customer dissatisfaction.

Key Code Samples

The four code samples below show the core functions from the pipeline. The full code is available in the GitHub repository.

Text Pre-processing Function

```
import re

def clean_text(val):
    val = str(val)
    val = val.lower()
    val = re.sub(r'^a-zA-Z\s', ' ', val)
    val = re.sub(r'\s+', ' ', val)
    val = val.strip()
    return val

df['clean_review'] = df['full_review'].apply(clean_text)
```

VADER Sentiment Classification

```
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer

analyzer = SentimentIntensityAnalyzer()

df['compound_score'] = df['clean_review'].apply(
    lambda x: analyzer.polarity_scores(x)['compound']
)

def classify_sentiment(score):
    if score >= 0.05:    return 'Positive'
    elif score <= -0.05: return 'Negative'
    else:               return 'Neutral'

df['sentiment'] = df['compound_score'].apply(classify_sentiment)
```

Multinomial Naive Bayes Classifier

```

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report

tfidf = TfidfVectorizer(stop_words='english')
X_vec = tfidf.fit_transform(df['clean_review'])

X_train, X_test, y_train, y_test = train_test_split(
    X_vec, df['sentiment'], test_size=0.2, random_state=42
)

model = MultinomialNB()
model.fit(X_train, y_train)
preds = model.predict(X_test)

print('Accuracy:', accuracy_score(y_test, preds))
print(classification_report(y_test, preds))

```

Keyword Frequency Analysis

```

from collections import Counter
from nltk.corpus import stopwords

stop_words = set(stopwords.words('english'))
extra      = {'sainsbury', 'sainsburys', 'tesco', 'always', 'time'}
all_stops  = stop_words.union(extra)

reviews = df[df['company_name'] == "Sainsbury's"]['clean_review']
text    = ' '.join(reviews)
words   = [w for w in text.split()
            if w not in all_stops and len(w) > 3]

top_15 = Counter(words).most_common(15)

```

Results

Sentiment Distribution

	Sainsburys	Tesco
Positive	68.1%	41.6%
Negative	26.5%	54.3%
Neutral	5.4%	4.1%
Total Reviews	1,000	1,000

Model Performance: Multinomial Naive Bayes

Class	Precision	Recall	F1-Score	Support
Negative	0.69	0.54	0.61	153
Neutral	0.00	0.00	0.00	27
Positive	0.68	0.87	0.76	220
Overall Accuracy	68.5%			400
Weighted F1	0.65			

The model performed well on positive and negative reviews but scored zero on neutral. This was expected because only 95 of the 2,000 reviews were neutral, representing 4.75 percent of the dataset. The classifier simply did not have enough examples to learn that class.

Top Negative Keywords Mapped to SERVQUAL Dimensions

Keyword	Frequency	SERVQUAL Dimension
delivery	361	Reliability
service	364	Responsiveness
customer	336	Responsiveness
staff	332	Assurance and Empathy
order	261	Reliability
time	246	Responsiveness
items	208	Reliability
store	357	Tangibles

Key Findings

- Sainsburys achieved a positive sentiment rate of 68.1 percent compared to only 41.6 percent for Tesco, a gap of 26.5 percentage points.

- Tesco's negative sentiment rate of 54.3 percent means more than half of all Tesco reviews were negative. This is a significant commercial risk because review valence is the strongest predictor of purchase intention.
- Delivery, service and staff were the three biggest drivers of negative sentiment. The word delivery appeared 361 times in negative reviews alone.
- The keyword findings map clearly onto the SERVQUAL Reliability and Responsiveness dimensions, both of which Tesco is failing on at scale.
- Sainsburys customers frequently mention helpful and staff in positive reviews, suggesting the Assurance and Empathy dimensions are a genuine competitive advantage.
- The Nectar loyalty programme appeared 186 times in positive Sainsburys reviews, suggesting some of the positive sentiment may be driven by reciprocity effects rather than pure service quality.

What I Learned From This Project

Before this project I had never built anything with Python beyond basic scripts. Getting the full pipeline working took a lot of trial and error. The most difficult part was understanding the difference between lexicon-based methods like VADER and supervised machine learning, and knowing when to use which approach.

Working with real, messy data was completely different from anything I had done in coursework. Reviews are noisy, inconsistent and sometimes sarcastic, and the model reflects that. A 68.5 percent accuracy rate is not perfect but it is honest, and understanding why it is not higher is as valuable as the number itself.

If I were doing this again I would implement LDA topic modelling earlier in the project and test multiple classifiers rather than committing to Naive Bayes from the start. I would also use a reference management tool from day one rather than tracking 48 sources manually.

Academic Context

Degree	BSc (Hons) Business Technology, First Class Honours
University	University of the West of Scotland
Module	COMP10034 Computing Honours Project
Submitted	April 2026
Supervisor	Shahriar Al-Ahmed
Dataset	2,000 Trustpilot reviews for Tesco and Sainsburys (November 2025 to February 2026)
Frameworks	SERVQUAL (Parasuraman et al., 1988) and eWOM Theory (Donthu et al., 2021)
References	48 peer-reviewed sources spanning 2009 to 2026